Assignment Set 2

CS-423 (Artificial Intelligence)

PART -I (Total Marks 1*11=11)

1) If p and q are Boolean variables, which of the following is a tautology (i.e., always correct irrespective of specific values of variables)?

- [ ] $(p \vee q) \rightarrow q$
- [ ] $p \vee (q \rightarrow p)$
- [x] $p \vee (p \rightarrow q)$
- [x] $p \rightarrow (q \rightarrow p)$

2) Which of the following propositions is logically equivalent to $A \vee (A \wedge B) \vee (A \wedge B \wedge C) \vee (A \wedge B \wedge C \wedge D) \vee (A \wedge B \wedge C \wedge D \wedge E)$?

- [x] A
- [ ] $A \wedge B \wedge C \wedge D \wedge E$
- [ ] E
- [ ] $A \vee B \vee C \vee D \vee E$

3) You have twice as many red balls as blue balls. If the probability of drawing a matched pair (same color) of balls is the same as the probability of drawing a mismatched pair (different color), then how many balls do you have in total? (assume that you have at least one ball and both the drawings are without replacement)

9

4) There are two boxes containing some black and green balls. The first box has 3 black and 1 green ball while the second box has 1 black and 3 green balls. One ball is picked from the first box randomly and put in the second box. Now, a ball is picked from the second box. What is the probability of it being a black ball?

0.35

5) There is a box with 9 white balls and 1 black ball. In each turn, we take out one ball. If it is black, we stop. If it is white, we throw the ball back in the box and repeat. What is the expected number of turns until we stop?

10

6) The ReLU function is a commonly used function in Machine Learning. The function is defined as follows: f(x) = max(0, x). Which of the following are correct numpy commands, if the function were to be applied on each element of a numpy array X of type float?

- [ ] numpy.abs(X) - X
- [x] numpy.multiply(X, X > 0)
- [ ] numpy.dot(numpy.transpose(X), X > 0)
- [x] (numpy.abs(X) + X)/2

7) Which of the following data structures is the most efficient in implementing a priority queue:

- ( ) Array
- (x) Heap
- ( ) Linked List
- ( ) Binary Search Tree

Given a weighted directed connected graph G, do the following operations change the optimal least-cost path from the source to the end vertex:Yes /No?

8) Adding a weight 10 to each edge's weight: ?

yes

9) Multiplying a weight 10 with each edge's weight:

no

10) What are the appropriate data structures for implementing breadth-first search and depth-first search algorithms respectively:

- ( ) Stack, Queue
- (x) Queue, Stack
- ( ) Stack, Heap
- ( ) Heap, Queue

11) Suppose that there are two nodes u and v in a graph. We find two paths connecting the two nodes using bfs and dfs. What can be said about the length of the two paths returned by bfs and dfs?

- ○ Path returned by bfs will always be shorter than that returned by dfs
- ○ Path returned by dfs will always be shorter than that returned by bfs
- ○ Both algorithms will always return the same path
- ✓ Path returned by bfs will never be longer than that returned by dfs